Background Information

Suppose we have yield data from a corn field. When the corn seed was planted, it was planted at different seed densities in different parts of the field. The seeding density was targeted to be higher in the parts of the field that were expected to be more fertile. How is fertility determined? We can estimate fertility based on the past years yield over different parts of the field.

In one field, corn was planted in 2018, following soybeans in 2017. Some parts of the field had high corn yields, some parts of the field had low corn yields. Some parts of the field were planted with more seed, some parts were planted with less seed. But the parts of the field planted with low seed in 2018 were also low yielding in 2017; that's why they were planted with less seed.

So here is a dilemma. If a specific region of the corn field was low yielding in 2018, was it because it was seeded at a lower rate, or because it is just a poor part of the field, as indicated by yield 2017?

To study this, I need to combine individual yield and seeding maps (data files with yield and seeding information tagged with a GPS position). Your task is to start with 6 files, aggregate the data by first creating a set of grid cells, then merging the data by grid cell into a single data set for further analysis. The data files, crop, year and data column to be aggregated are listed below.

File Name	Crop	Year	Original Variable	Aggregated Variable
`A 2017 Soybeans Harvest.csv`	Soybean	2017	`Yield`	`Y17`
`A 2018 Corn Seeding.csv`	Corn	2018	`AppliedRate`	`AR18`
`A 2018 Corn Harvest.csv`	Corn	2018	`Yield`	`Y18`
`A 2019 Soybeans Harvest.csv`	Soybean	2019	`Yield`	`Y19`
`A 2020 Corn Seeding.csv`	Corn	2020	`AppliedRate`	`AR20`
`A 2020 Corn Harvest.csv`	Corn	2020	`Yield`	`Y20`

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``{python mergDtaPlot}

import seaborn as sns
import matplotlib.pyplot as plt
from scipy import stats

# Scatter plot with regression line
sns.lmplot(x="Weight_x", y="Weight_y", data=merged, aspect=1.5)
plt.title("College vs High School Weight")
plt.xlabel("High School Weight (Weight_x)")
plt.ylabel("College Weight (Weight_y)")
plt.show()

# Regression analysis
slope, intercept, r_value, p_value, std_err = stats.linregress(merged["Weight_x"], merged["Weight_y"])
print(f"Slope: {slope:.2f}")
print(f"Intercept: {intercept:.2f}")
print(f"R-squared: {r_value**2:.3f}")
print(f"P-value: {p_value:.5f}")
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# Clean data
merged["Weight_x"] = (
    merged["Weight_x"].astype(str)
                  .str.replace(r"[^\d.]", "", regex=True)
                  .astype(float)
)
merged["Weight_y"] = (
    merged["Weight_y"].astype(str)
                  .str.replace(r"[^\d.]", "", regex=True)
                  .astype(float)
)
```